

CROP DISEASE EXPERT SYSTEM

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===== DIAGNOSIS RESULT =====

Disease : **Bacterial Spot**
Certainty : **0.86 (86.0%)**

===== EXTRACTED FEATURES =====

Lesion Area % : 18.73
Mean Hue (degrees) : 88.62
Mean Saturation : 162.48
Mean Value : 98.37
Lesion Count : 8
Shape Compactness : 0.412
GLCM Contrast : 0.184
GLCM Homogeneity : 0.934

===== SYMBOLIC FACTS =====

Lesion Area : Medium (5-30%)
Lesion Color : Dark
Lesion Count : High (>=3)
Lesion Shape : Circular/Irregular
Texture : Smooth to Slightly Homogeneous

===== RULES FIRED =====

R6: Multiple small lesions -> Bacterial Spot
R8: Dark small lesions with high count -> Bacterial Spot
R9: Medium lesion area with high count -> Bacterial Spot

Overall CF Score for Bacterial Spot : **0.86 (86.0%)**

Visual Studio Code interface showing the project structure and terminal output.

Explorer:

- Crop_Disease_Expert_Syst...
- _pycache_
- venv
- image_processing.py
- main.py
- README.txt
- requirements.txt
- rule_engine.py

Terminal:

```
python -m pip install --upgrade pip
Successfully installed pillow-10.4.0 imageio-2.37.4 lazy-loader-0.5 networkx-3.6.1 numpy-2.2.5 opencv-python-4.10.0.84 packaging-26.3 scikit-image-0.25.0 scipy-1.14.1 tifffile-2025.12.1
[notice] A new release of pip is available: 25.2 -> 26.2.1
[notice] To update, run: c:\Users\Wen\Downloads\Crop_Disease_Expert_System_VS_Code\venv\Scripts\python.exe -m pip install --upgrade pip
PS C:\Users\Wen\Downloads\Crop_Disease_Expert_System_VS_Code> .\venv\Scripts\python.exe main.py
```

Chat:

Build with Agent
AI responses may be inaccurate
Generate Agent
Instructions to onboard AI onto your codebase.